

Gabarito - 2ª Av. - 3ª Chamada - Prob. Est. - Comp.

2) $X \sim \text{Bin}(10; 0,7)$

2.1) $P[X=5] = 0,1029$

2.2) $P[X \geq 5] = 1 - P[X \leq 4] = 1 - 0,0474 = 0,9526$

2.3) $P[X \geq 50] = P[50 \leq X \leq 100] = P[49,5 \leq X \leq 100,5]$

$\mu = 70$
 $\sigma = 4,5826$

$$\approx P\left[\frac{49,5-70}{4,5826} \leq Z \leq \frac{100,5-70}{4,5826}\right] = \Phi(6,66) - \Phi(-4,47)$$

$$= 1 - 0 = 1$$

2) $X = \text{"Núm. de meninas"} \Rightarrow X \sim \text{BN}(2; \frac{1}{2})$

2.1) $E[X] = n \cdot \frac{1-p}{p} = 2 \cdot \frac{\frac{1}{2}}{\frac{1}{2}} = 2$

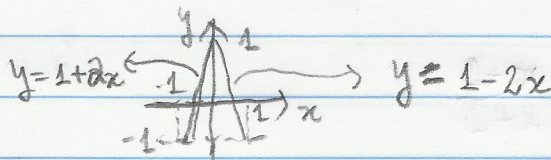
2.2) $P[X \geq 2] = 1 - P[X=0] - P[X=1] = 1 - \left[\binom{2}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^2 + \binom{2}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^1 \right]$

$$= 1 - \frac{1}{2^2} \left[1 + 2 \cdot \frac{1}{2} \right] = 1 - \frac{1}{4} [1 + 1] = 1 - \frac{1}{4} \cdot 2 = \frac{1}{2}$$

3) 3.1) $E[X] = e^{\mu + \frac{\sigma^2}{2}} = 22,7599$

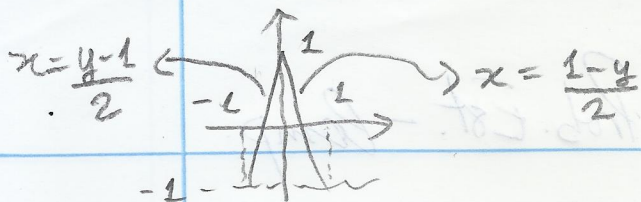
$$V(X) = \mu^2 [e^{\sigma^2} - 1] = 147,1288 \Rightarrow \text{DP}[X] = \sqrt{V(X)} = 12,1297$$

3.2) $P[X < 50] = P[\log X < \log 50] = P\left[Z < \frac{\log 50 - 3}{0,5}\right] = P[Z < 1,82]$
 $= \Phi(1,82) = 0,9656$



4) 4.1) Área do triângulo = $\frac{2 \cdot 2}{2} = 2$. Altura da fdp = K . Logo, $1 = 2 \cdot K$. Assim, $f(x, y) = 1/2$, $-1 < y < 1 - 2|x|$, $-1 < x < 1$.

$$\text{Logo, } f_X(x) = \int_{-1}^{1-2|x|} \frac{1}{2} dy = \frac{1}{2} [1 - 2|x| - (-1)] = 1 - |x|, \quad -1 < x < 1.$$



Logo

$$f_Y(y) = \int_{\frac{y-1}{2}}^{\frac{1-y}{2}} \frac{1}{2} dx = \frac{1}{2} \cdot \frac{1}{2} [1-y - (y-1)] = \frac{1-y}{2}, \quad -1 < y < 1.$$

$$\begin{aligned} \text{4.2} \quad E[XY] &= \int_{-1}^1 \int_{\frac{y-1}{2}}^{\frac{1-y}{2}} xy \, dx \, dy = \int_{-1}^1 y \left[\frac{x^2}{2} \Big|_{\frac{y-1}{2}}^{\frac{1-y}{2}} \right] dy \\ &= \int_{-1}^1 y \left[\frac{(1-y)^2}{4} - \frac{(-1)(1-y)^2}{4} \right] dy = 0 \end{aligned}$$

$$\begin{aligned} E[X] &= \int_{-1}^1 x(1-|x|) dx = \int_{-1}^0 x(1+x) dx + \int_0^1 x(1-x) dx \\ &= \left[\frac{x^2}{2} + \frac{x^3}{3} \Big|_{-1}^0 \right] + \left[\frac{x^2}{2} - \frac{x^3}{3} \Big|_0^1 \right] = -\frac{1}{2} + \frac{1}{3} + \frac{1}{2} - \frac{1}{3} = 0 \end{aligned}$$

Logo

$$\text{Cov}(X, Y) = 0 - 0 \cdot E[Y] = 0$$

$$\text{4.3} \quad X \text{ e } Y \text{ não são indep., pois } f(x, y) \neq f_X(x)f_Y(y).$$